

Tanay Agrawal

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EDUCATION

University of Southern California (USC)

Los Angeles, CA

M.S. Electrical Engineering | Research Assistant, WiDeS Group | GPA: 3.75/4

Aug 2025 – Aug 2027

• Coursework: EE 535 (Wireless Communications), EE 559 (Machine Learning), EE 510 (Linear Algebra), EE 503 (Probability)

BITS Pilani K K Birla Goa Campus

Goa, India

B.E. Electronics and Communication Engineering | GPA: 8.32/10

Aug 2021 – Aug 2025

• Bachelor's Thesis: *Linear Channel Modeling and Linear Detectors for Multi-Mode Fibers* | TU Munich

SKILLS

Lab Instruments: VNA, Anechoic Chamber, AWG, Oscilloscopes, Signal Analyzers, Phased Arrays, GPSDO, IF-over-Fiber Systems

Programming: Python, MATLAB, C/C++

Software: Simulink, STM32 Cube IDE, ESP-IDF, CST Studio Suite, GNU Radio, Vivado (Model-based design)

Protocols: 802.11 (WiFi), BLE (NimBLE), SPI, UART, I2C

Hardware: PCB design (KiCAD/OrCAD) through fabrication and assembly

EXPERIENCE

USC Viterbi | WiDeS Group

Los Angeles, CA

Research Assistant under Prof. Andreas Molisch

Aug 2025 – Present

- **RFSoc FMCW Radar (DARPA funded) – Lead Engineer:** Designed and programmed a complete channel sounder at 6.5 GHz using Zynq UltraScale+ RFSoc FPGA, GPSDO-disciplined timing, PA, LNA, BPF, including RF chain characterization, timing and stability testing, waveform generation. Integrated custom MIMO antenna array; performed SOLT-calibrated VNA characterization of full RF chain individually and cascaded.
- **Distributed MIMO Channel Sounder Infrastructure – 2IC:** Lead the full outdoor deployment of a coherent distributed MIMO channel sounder at 28 GHz. Integrated GPSDO-disciplined timing, 2-stage LO chain, IF-over-fiber interconnects, and PPS/10 MHz reference distribution across three transmitter carts and a receiver cart for extensive outdoor urban mmWave measurements. Contributed to STM32 MCU firmware and full array calibration in anechoic chamber.
- **PDP MPC Extraction for Cell-Free Massive MIMO:** Designed a custom multi-stage MATLAB pipeline for 3–4 GHz channel data using 2D image-based peak detection, DBSCAN clustering, and constrained least-squares solver. Recovered per-MPC delay profiles from scalar PDP data alone, enabling multipath correlation analysis for cell-free massive MIMO without angular information.

BITS Pilani | Lead Researcher

Goa, India

Guided by Prof. Ravi Kadlimatti – 3 first-author IEEE publications

Jan 2024 – Aug 2025

- **Led the research on OTFS modulation scheme for the Department of Electrical and Electronics Engineering at BITS Pilani** | [GitHub - Code-Aided](#) | [GitHub - UAC](#)
- **Code-Aided Delay-Doppler Domain Channel Estimation:** Novel contribution in proposing P3 polyphase coded pilot scheme for embedded pilot-aided OTFS channel estimation achieving 6 dB PAPR reduction while converging to ideal BER performance. *First author. Published: IEEE CCNC 2026, IEEE WCNC 2026.*
- **Equalization Techniques for OTFS in UWA Environments:** Implemented a statistically equivalent UWA channel model for OTFS and compared LMMSE, MRC, and MPA detectors across three depth scenarios ; achieved BER 10^{-3} at 25 dB SNR. *First author. Published: APWiMob 2025.*

PROJECTS

ESP32 BLE Scanner with Anechoic Chamber Validation | C++, Python, Matlab | [GitHub](#)

Jun 2026

- Built dual-core BLE scanner using NimBLE radio stack on ESP32 WROOM 32E. Validated MATLAB-based RSSI measurements with Python CSV logging pipeline against theoretical path loss model in anechoic chamber.

Drone RF Signal Classification | Python, NumPy, scikit-learn

March 2026 – May 2026

- Engineered 89-feature IQ signal pipeline (time-domain, spectral, phase, burst statistics) from 17,000+ raw captures with hierarchical SVM + MLP ensemble achieved 66.9% balanced accuracy across -12 to $+30$ dB SNR.

Scalable Ku/K/C-Band Microstrip Patch Antenna Array | CST Studio Suite

Apr 2024 – Jul 2024

- Designed scalable RT-duroid 5880LZ patch array with coaxial probe feed achieving consistent 3 dBi directivity increase per element doubling (11.47 to 26.36 dBi, 1 to 32 elements); validated across Ku (13.97 GHz), K (22.66 GHz), and C (7.6 GHz) bands via linear dimension scaling. *Submitted: PIER Journal.*

PUBLICATIONS

- [1] T. Agrawal and R. Kadlimatti, “Low-PAPR Channel Estimation for MIMO-OTFS Systems via Embedded Pilot Codes,” *2026 IEEE Wireless Communications and Networking Conference (WCNC)*, Kuala Lumpur, Malaysia, 2026.
- [2] T. Agrawal and R. Kadlimatti, “Embedded Pilot Code Aided Delay Doppler Domain Channel Estimation,” *2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC)*, Las Vegas, NV, USA, 2026.
- [3] T. Agrawal, R. Kadlimatti and N. Gupta, “Comparison of Channel Equalization Methods for an OTFS-Based Underwater Acoustic Communication System,” *2025 IEEE Asia Pacific Conference on Wireless and Mobile (APWiMob)*, Bali, Indonesia, 2025.